A Linear-Scaling Fluid-Dynamic Vector Model for Gravitational Trajectory Tracking

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Abstract / Executive Summary

This paper describes a novel framework to configure a gravitational model designed to track gravitational movement and celestial objects without the use of complex tensor calculus. By utilizing established fluid dynamics laws and modified electromagnetic monopole modeling, gravitational interference, trajectories, and orbits can be calculated using a linear-scaling (O(n)) vector architecture.

Core Axioms and Operational Rules

1. The Spatial Substrate

The vacuum of space is treated as a continuous fluid medium governed by standard fluid-dynamic principles.

2. Theoretical Foundation

The system utilizes proven, established formulas from fluid dynamics and electromagnetic monopole field theory, requiring no modifications to foundational physical laws.

3. Gravitational Monopole Field Configuration

Gravity is modeled as an electromagnetic monopole field operating under three distinct, hardcoded architectural rules:

- **Rule A: The 2=1 Energy Scaling Law**
  - **Natural Inverse-Square Law:** The 2=1 energy-to-matter count naturally scales within 3D space to establish the baseline inverse-square law (1/r^2).
  - **The Structural Lower Bound:** The model enforces a hard physical floor at 1/2 of a whole unit, where matter reality interactions break down.
  - **Error Prevention:** This floor prevents the system distance variable from ever reaching absolute zero, completely eliminating divide-by-zero crashes.
- **Rule B: Vector Inversion and Self-Attraction**
  - **Strict Inward Attractiveness:** The directional signs of the monopole vectors are inverted so that the field is strictly attractive.
  - **Fluid Stabilization:** This single directional flow stabilizes the fluid field, preventing non-physical shockwaves and unnatural wake effects in the spatial medium.
  - **Field Stacking:** Because gravity tracks gravity, the field lines attract to themselves, establishing a compounding force gradient that drops off at a 1/r^3 geometric rate.
  - **Orbital Stabilization:** This localized 1/r^3 self-attraction gradient stabilizes macro-structures, providing the extra gravitational binding energy required to

    prevent the outermost stars of a galaxy from flying off without requiring theoretical dark matter.
- **Rule C: Environmental and Object Mapping**
    - All physical characteristics of tracked bodies—including mass, velocity, macroscopic rotation/spin, thermal output, and geometric shapes—are introduced into the running fluid ledger as localized interference variables and vector patterns.

4. Conservation of Physical Reality

All external physical laws, including thermodynamics, fluid continuity, and standard electromagnetism, are left whole and entirely unchanged. This preserves the structural integrity of real-world interactions while simplifying the underlying computational framework.

5. Testing and Results

I built a python simulator to run elliptic orbital calculations on my laptop as a proof of math and proof of concept.

50,000,000 steps without drift, loss, and kept its elliptical orbit and displayed the wobble nearly identically to real world observations. Through all 50 Million cycles.

The python program and
abridged python output below

--- RUNNING STABLE ORBITAL PREDICTION (50,000,000 DAYS) ---
Day 00 -> Pos X: 45,760,576,784.73 | Pos Y: 5,097,600,000.00 | Speed: 58,952.8 m/s
Day 01 -> Pos X: 45,046,161,924.85 | Pos Y: 10,142,287,262.06 | Speed: 58,812.2 m/s
Day 02 -> Pos X: 43,868,143,050.67 | Pos Y: 15,082,592,599.26 | Speed: 58,581.3 m/s
Day 03 -> Pos X: 42,244,928,368.50 | Pos Y: 19,869,832,259.83 | Speed: 58,265.1 m/s
Day 04 -> Pos X: 40,201,149,435.23 | Pos Y: 24,459,260,219.76 | Speed: 57,870.0 m/s
Day 05 -> Pos X: 37,766,653,599.01 | Pos Y: 28,810,967,454.40 | Speed: 57,403.9 m/s
Day 06 -> Pos X: 34,975,372,895.82 | Pos Y: 32,890,494,969.34 | Speed: 56,875.3 m/s
Day 07 -> Pos X: 31,864,156,046.95 | Pos Y: 36,669,157,695.17 | Speed: 56,293.2 m/s
Day 08 -> Pos X: 28,471,639,796.00 | Pos Y: 40,124,102,067.41 | Speed: 55,666.8 m/s
Day 09 -> Pos X: 24,837,218,727.10 | Pos Y: 43,238,138,305.26 | Speed: 55,004.9 m/s
Day 10 -> Pos X: 21,000,152,866.15 | Pos Y: 45,999,398,185.90 | Speed: 54,316.2 m/s
Day 15 -> Pos X: 60,504,691.03 | Pos Y: 54,408,104,821.90 | Speed: 50,723.5 m/s
Day 20 -> Pos X: -21,005,425,264.67 | Pos Y: 54,375,781,545.48 | Speed: 47,351.1 m/s
Day 25 -> Pos X: -39,487,419,600.26 | Pos Y: 47,436,972,779.48 | Speed: 44,551.7 m/s
Day 30 -> Pos X: -53,795,153,928.71 | Pos Y: 35,406,750,189.25 | Speed: 42,461.4 m/s
Day 35 -> Pos X: -63,066,301,317.55 | Pos Y: 20,026,997,900.35 | Speed: 41,122.8 m/s
Day 40 -> Pos X: -66,880,889,863.51 | Pos Y: 2,899,392,631.14 | Speed: 40,546.1 m/s
Day 45 -> Pos X: -65,103,108,597.19 | Pos Y: -14,473,443,468.53 | Speed: 40,732.7 m/s
Day 50 -> Pos X: -57,821,478,087.51 | Pos Y: -30,614,229,979.13 | Speed: 41,682.3 m/s
Day 55 -> Pos X: -45,370,093,427.85 | Pos Y: -43,994,132,591.59 | Speed: 43,390.8 m/s
Day 60 -> Pos X: -28,435,984,002.61 | Pos Y: -52,974,541,655.35 | Speed: 45,836.8 m/s
Day 65 -> Pos X: -8,276,388,946.55 | Pos Y: -55,814,654,136.41 | Speed: 48,942.9 m/s
Day 70 -> Pos X: 12,940,970,447.18 | Pos Y: -50,864,131,264.19 | Speed: 52,487.1 m/s
Day 75 -> Pos X: 31,795,135,831.76 | Pos Y: -37,175,807,444.58 | Speed: 55,949.9 m/s
Day 80 -> Pos X: 43,807,682,347.97 | Pos Y: -15,721,454,006.79 | Speed: 58,415.7 m/s
Day 85 -> Pos X: 45,105,242,122.48 | Pos Y: 9,480,799,671.91 | Speed: 58,901.7 m/s
Day 90 -> Pos X: 35,085,692,116.00 | Pos Y: 32,345,142,590.25 | Speed: 57,176.3 m/s
Day 95 -> Pos X: 17,002,940,696.29 | Pos Y: 47,963,912,601.48 | Speed: 54,003.8 m/s
Day 100 -> Pos X: -4,428,094,309.33 | Pos Y: 54,581,149,555.94 | Speed: 50,407.5 m/s
Day 105 -> Pos X: -25,298,002,982.12 | Pos Y: 52,880,825,312.36 | Speed: 47,076.1 m/s
Day 110 -> Pos X: -43,088,920,619.82 | Pos Y: 44,569,769,665.40 | Speed: 44,336.3 m/s
Day 115 -> Pos X: -56,377,394,568.63 | Pos Y: 31,533,896,512.81 | Speed: 42,311.8 m/s
Day 120 -> Pos X: -64,432,200,752.53 | Pos Y: 15,539,253,646.18 | Speed: 41,040.7 m/s
Day 125 -> Pos X: -66,938,625,617.67 | Pos Y: -1,807,164,426.25 | Speed: 40,531.8 m/s
Day 130 -> Pos X: -63,852,529,305.86 | Pos Y: -19,004,418,029.26 | Speed: 40,786.2 m/s
Day 135 -> Pos X: -55,351,685,106.74 | Pos Y: -34,580,395,661.72 | Speed: 41,803.4 m/s
Day 140 -> Pos X: -41,867,692,929.39 | Pos Y: -47,016,987,274.22 | Speed: 43,578.8 m/s
Day 145 -> Pos X: -24,204,413,554.11 | Pos Y: -54,700,650,926.86 | Speed: 46,087.9 m/s
Day 150 -> Pos X: -3,765,220,018.11 | Pos Y: -55,950,817,890.78 | Speed: 49,245.1 m/s
Day 155 -> Pos X: 17,108,595,154.29 | Pos Y: -49,254,872,311.88 | Speed: 52,808.7 m/s
Day 160 -> Pos X: 34,843,843,583.26 | Pos Y: -33,946,246,197.88 | Speed: 56,225.3 m/s
Day 165 -> Pos X: 44,983,854,373.79 | Pos Y: -11,447,670,053.98 | Speed: 58,550.1 m/s
Day 170 -> Pos X: 44,048,436,786.28 | Pos Y: 13,784,965,273.73 | Speed: 58,832.4 m/s
Day 175 -> Pos X: 32,097,003,507.33 | Pos Y: 35,624,379,586.69 | Speed: 56,937.1 m/s
Day 180 -> Pos X: 12,849,120,464.34 | Pos Y: 49,578,100,362.83 | Speed: 53,688.3 m/s

Day 185 -> Pos X: -8,920,054,643.73 | Pos Y: 54,385,900,277.35 | Speed: 50,093.9 m/s
Day 190 -> Pos X: -29,465,091,890.92 | Pos Y: 51,052,356,398.94 | Speed: 46,806.0 m/s
Day 195 -> Pos X: -46,467,265,487.03 | Pos Y: 41,435,396,408.55 | Speed: 44,126.6 m/s
Day 200 -> Pos X: -58,669,576,395.65 | Pos Y: 27,478,103,013.83 | Speed: 42,168.1 m/s
Day 205 -> Pos X: -65,469,989,535.52 | Pos Y: 10,962,137,522.15 | Speed: 40,964.6 m/s
Day 210 -> Pos X: -66,657,449,923.65 | Pos Y: -6,506,006,977.37 | Speed: 40,523.5 m/s
Day 215 -> Pos X: -62,278,640,414.46 | Pos Y: -23,431,716,271.89 | Speed: 40,845.7 m/s
Day 220 -> Pos X: -52,600,528,491.33 | Pos Y: -38,352,585,022.31 | Speed: 41,930.6 m/s
Day 225 -> Pos X: -38,152,829,066.85 | Pos Y: -49,766,654,504.05 | Speed: 43,772.6 m/s
Day 230 -> Pos X: -19,857,080,673.43 | Pos Y: -56,093,064,055.26 | Speed: 46,344.2 m/s
Day 235 -> Pos X: 737,577,621.17 | Pos Y: -55,723,194,008.85 | Speed: 49,550.7 m/s
Day 240 -> Pos X: 21,122,211,880.17 | Pos Y: -47,300,064,961.71 | Speed: 53,129.5 m/s
Day 245 -> Pos X: 37,592,951,840.47 | Pos Y: -30,459,841,725.40 | Speed: 56,492.2 m/s
Day 250 -> Pos X: 45,762,848,220.31 | Pos Y: -7,084,225,847.35 | Speed: 58,667.8 m/s
Day 255 -> Pos X: 42,599,526,935.66 | Pos Y: 17,973,570,103.42 | Speed: 58,745.2 m/s
Day 260 -> Pos X: 28,825,550,233.81 | Pos Y: 38,623,049,969.09 | Speed: 56,687.2 m/s
Day 265 -> Pos X: 8,569,662,466.65 | Pos Y: 50,831,246,552.32 | Speed: 53,370.3 m/s
Day 270 -> Pos X: -13,385,494,773.71 | Pos Y: 53,824,521,247.63 | Speed: 49,783.0 m/s
Day 275 -> Pos X: -33,481,421,258.25 | Pos Y: 48,901,976,759.48 | Speed: 46,540.8 m/s
Day 280 -> Pos X: -49,603,301,685.79 | Pos Y: 38,051,889,941.54 | Speed: 43,922.6 m/s
Day 285 -> Pos X: -60,659,276,140.08 | Pos Y: 23,261,456,929.24 | Speed: 42,030.4 m/s
Day 290 -> Pos X: -66,174,186,560.94 | Pos Y: 6,319,828,353.64 | Speed: 40,894.5 m/s
Day 295 -> Pos X: -66,038,830,719.60 | Pos Y: -11,172,587,341.05 | Speed: 40,521.2 m/s
Day 300 -> Pos X: -60,389,721,848.69 | Pos Y: -27,732,053,822.78 | Speed: 40,911.2 m/s
Day 305 -> Pos X: -49,582,814,608.08 | Pos Y: -41,910,441,441.10 | Speed: 42,063.8 m/s
Day 310 -> Pos X: -34,246,332,886.03 | Pos Y: -52,227,529,051.25 | Speed: 43,972.3 m/s
Day 315 -> Pos X: -15,419,942,733.08 | Pos Y: -57,143,057,338.22 | Speed: 46,605.7 m/s
Day 320 -> Pos X: 5,202,644,819.28 | Pos Y: -55,132,504,199.86 | Speed: 49,859.4 m/s
Day 325 -> Pos X: 24,952,530,231.19 | Pos Y: -45,012,684,051.53 | Speed: 53,448.9 m/s
Day 330 -> Pos X: 40,019,763,974.99 | Pos Y: -26,743,341,223.71 | Speed: 56,749.7 m/s
Day 335 -> Pos X: 46,137,452,427.47 | Pos Y: -2,667,841,588.07 | Speed: 58,768.3 m/s
Day 340 -> Pos X: 40,771,267,528.08 | Pos Y: 22,011,221,311.63 | Speed: 58,640.4 m/s
Day 345 -> Pos X: 25,298,331,546.64 | Pos Y: 41,318,059,208.31 | Speed: 56,427.3 m/s
Day 350 -> Pos X: 4,196,076,402.13 | Pos Y: 51,715,382,493.65 | Speed: 53,050.6 m/s
Day 355 -> Pos X: -17,795,021,515.06 | Pos Y: 52,901,541,188.72 | Speed: 49,475.0 m/s
Day 360 -> Pos X: -37,322,827,037.65 | Pos Y: 46,443,123,390.23 | Speed: 46,280.6 m/s
Day 365 -> Pos X: -52,479,348,168.16 | Pos Y: 34,438,589,065.21 | Speed: 43,724.3 m/s
Day 370 -> Pos X: -62,335,747,025.13 | Pos Y: 18,906,836,903.22 | Speed: 41,898.7 m/s
Day 375 -> Pos X: -66,541,077,814.75 | Pos Y: 1,636,807,069.18 | Speed: 40,830.5 m/s
Day 380 -> Pos X: -65,085,999,120.85 | Pos Y: -15,782,529,400.36 | Speed: 40,525.0 m/s
Day 385 -> Pos X: -58,195,724,071.53 | Pos Y: -31,882,780,725.21 | Speed: 40,982.8 m/s
Day 390 -> Pos X: -46,314,833,987.67 | Pos Y: -45,234,698,323.28 | Speed: 42,203.0 m/s
Day 395 -> Pos X: -30,170,221,452.54 | Pos Y: -54,385,545,432.70 | Speed: 44,177.7 m/s
Day 400 -> Pos X: -10,919,693,464.89 | Pos Y: -57,843,884,146.74 | Speed: 46,872.1 m/s
Day 405 -> Pos X: 9,600,564,934.37 | Pos Y: -54,181,834,322.07 | Speed: 50,171.0 m/s
Day 410 -> Pos X: 28,571,228,069.52 | Pos Y: -42,408,258,871.35 | Speed: 53,766.3 m/s

Day 415 -> Pos X: 42,103,933,320.45 | Pos Y: -22,825,637,466.45 | Speed: 56,997.1 m/s
Day 420 -> Pos X: 46,103,941,341.28 | Pos Y: 1,764,125,650.33 | Speed: 58,851.2 m/s
Day 425 -> Pos X: 38,579,657,205.34 | Pos Y: 25,863,987,062.28 | Speed: 58,518.5 m/s
Day 430 -> Pos X: 21,544,115,546.10 | Pos Y: 43,688,993,875.32 | Speed: 56,158.2 m/s
Day 435 -> Pos X: -239,829,295.26 | Pos Y: 52,225,283,584.71 | Speed: 52,729.5 m/s
Day 440 -> Pos X: -22,119,907,482.54 | Pos Y: 51,623,779,818.56 | Speed: 49,170.3 m/s
Day 445 -> Pos X: -40,966,375,506.85 | Pos Y: 43,690,964,344.77 | Speed: 46,025.5 m/s
Day 450 -> Pos X: -55,079,276,530.75 | Pos Y: 30,616,016,386.86 | Speed: 43,531.9 m/s
Day 455 -> Pos X: -63,689,964,307.39 | Pos Y: 14,437,788,457.58 | Speed: 41,773.0 m/s
Day 460 -> Pos X: -66,568,729,933.48 | Pos Y: -3,062,269,518.02 | Speed: 40,772.4 m/s
Day 465 -> Pos X: -63,803,932,807.78 | Pos Y: -20,311,752,600.87 | Speed: 40,534.8 m/s
Day 470 -> Pos X: -55,708,221,877.01 | Pos Y: -35,861,999,424.69 | Speed: 41,060.4 m/s
Day 475 -> Pos X: -42,814,284,959.10 | Pos Y: -48,307,286,863.69 | Speed: 42,348.1 m/s
Day 480 -> Pos X: -25,947,589,271.73 | Pos Y: -56,228,269,984.79 | Speed: 44,388.9 m/s
Day 485 -> Pos X: -6,383,617,901.78 | Pos Y: -58,190,842,465.27 | Speed: 47,143.5 m/s
Day 490 -> Pos X: 13,902,051,676.11 | Pos Y: -52,876,659,420.60 | Speed: 50,485.2 m/s
Day 495 -> Pos X: 31,951,181,274.83 | Pos Y: -39,504,815,900.44 | Speed: 54,081.1 m/s

Day 49999500 -> Pos X: 50,075,011,154.18 | Pos Y: 38,226,005,976.63 | Speed: 43,546.3 m/s
Day 49999505 -> Pos X: 34,215,200,679.47 | Pos Y: 49,128,875,755.14 | Speed: 46,044.7 m/s
Day 49999510 -> Pos X: 14,410,941,057.30 | Pos Y: 54,255,443,126.24 | Speed: 49,193.4 m/s
Day 49999515 -> Pos X: -7,335,873,725.15 | Pos Y: 51,681,747,627.76 | Speed: 52,754.0 m/s
Day 49999520 -> Pos X: -27,660,405,040.27 | Pos Y: 40,072,228,835.78 | Speed: 56,179.0 m/s
Day 49999525 -> Pos X: -41,930,893,296.66 | Pos Y: 19,956,950,387.64 | Speed: 58,528.4 m/s
Day 49999530 -> Pos X: -45,873,266,262.85 | Pos Y: -4,981,422,700.06 | Speed: 58,845.5 m/s
Day 49999535 -> Pos X: -38,345,469,148.49 | Pos Y: -28,724,319,351.06 | Speed: 56,978.6 m/s
Day 49999540 -> Pos X: -22,121,540,524.44 | Pos Y: -46,129,993,353.41 | Speed: 53,742.2 m/s
Day 49999545 -> Pos X: -1,655,170,593.92 | Pos Y: -55,027,739,298.89 | Speed: 50,147.2 m/s
Day 49999550 -> Pos X: 19,169,481,241.21 | Pos Y: -55,682,899,969.18 | Speed: 46,851.6 m/s
Day 49999555 -> Pos X: 37,710,823,179.29 | Pos Y: -49,481,517,934.06 | Speed: 44,161.9 m/s
Day 49999560 -> Pos X: 52,360,110,282.86 | Pos Y: -38,098,509,217.70 | Speed: 42,192.2 m/s
Day 49999565 -> Pos X: 62,186,959,503.96 | Pos Y: -23,174,164,181.03 | Speed: 40,977.1 m/s
Day 49999570 -> Pos X: 66,680,678,347.23 | Pos Y: -6,249,206,063.15 | Speed: 40,524.5 m/s
Day 49999575 -> Pos X: 65,606,032,203.05 | Pos Y: 11,198,628,403.74 | Speed: 40,835.1 m/s
Day 49999580 -> Pos X: 58,950,403,907.77 | Pos Y: 27,683,523,857.20 | Speed: 41,908.6 m/s
Day 49999585 -> Pos X: 46,950,956,968.69 | Pos Y: 41,633,151,551.93 | Speed: 43,739.2 m/s
Day 49999590 -> Pos X: 30,212,587,198.98 | Pos Y: 51,325,530,676.06 | Speed: 46,300.2 m/s
Day 49999595 -> Pos X: 9,947,006,051.92 | Pos Y: 54,893,725,838.93 | Speed: 49,498.4 m/s
Day 49999600 -> Pos X: -11,650,682,156.28 | Pos Y: 50,533,899,430.39 | Speed: 53,075.0 m/s
Day 49999605 -> Pos X: -31,030,041,660.45 | Pos Y: 37,180,840,359.32 | Speed: 56,447.4 m/s
Day 49999610 -> Pos X: -43,538,270,263.31 | Pos Y: 15,826,194,168.27 | Speed: 58,649.0 m/s
Day 49999615 -> Pos X: -45,261,122,120.13 | Pos Y: -9,371,049,642.78 | Speed: 58,761.3 m/s
Day 49999620 -> Pos X: -35,713,731,052.52 | Pos Y: -32,297,386,927.57 | Speed: 56,730.5 m/s
Day 49999625 -> Pos X: -18,163,037,498.53 | Pos Y: -48,181,856,622.46 | Speed: 53,424.6 m/s
Day 49999630 -> Pos X: 2,834,390,391.77 | Pos Y: -55,331,624,063.72 | Speed: 49,835.8 m/s

Day 49999635 -> Pos X: 23,521,568,533.93 | Pos Y: -54,338,902,152.86 | Speed: 46,585.6 m/s
Day 49999640 -> Pos X: 41,433,099,821.43 | Pos Y: -46,757,584,231.97 | Speed: 43,956.9 m/s
Day 49999645 -> Pos X: 55,114,793,810.73 | Pos Y: -34,337,555,475.15 | Speed: 42,053.5 m/s
Day 49999650 -> Pos X: 63,759,963,293.53 | Pos Y: -18,750,443,778.26 | Speed: 40,906.0 m/s
Day 49999655 -> Pos X: 66,959,377,264.02 | Pos Y: -1,550,072,523.60 | Speed: 40,521.2 m/s
Day 49999660 -> Pos X: 64,568,173,482.28 | Pos Y: 15,779,764,022.58 | Speed: 40,899.6 m/s
Day 49999665 -> Pos X: 56,663,748,681.95 | Pos Y: 31,750,698,838.71 | Speed: 42,040.7 m/s
Day 49999670 -> Pos X: 43,583,849,335.13 | Pos Y: 44,793,339,467.05 | Speed: 43,937.9 m/s
Day 49999675 -> Pos X: 26,056,546,037.46 | Pos Y: 53,203,314,201.35 | Speed: 46,560.8 m/s
Day 49999680 -> Pos X: 5,450,659,959.04 | Pos Y: 55,168,625,792.26 | Speed: 49,806.6 m/s
Day 49999685 -> Pos X: -15,849,523,739.67 | Pos Y: 49,024,706,736.31 | Speed: 53,394.6 m/s
Day 49999690 -> Pos X: -34,127,918,056.62 | Pos Y: 34,001,734,873.19 | Speed: 56,706.6 m/s
Day 49999695 -> Pos X: -44,759,206,062.65 | Pos Y: 11,565,401,382.11 | Speed: 58,752.5 m/s
Day 49999700 -> Pos X: -44,247,519,239.49 | Pos Y: -13,685,850,081.19 | Speed: 58,659.5 m/s
Day 49999705 -> Pos X: -32,772,318,404.46 | Pos Y: -35,621,641,101.28 | Speed: 56,472.2 m/s
Day 49999710 -> Pos X: -14,041,188,777.18 | Pos Y: -49,889,431,522.91 | Speed: 53,105.1 m/s
Day 49999715 -> Pos X: 7,338,278,828.42 | Pos Y: -55,269,690,279.83 | Speed: 49,527.3 m/s
Day 49999720 -> Pos X: 27,760,768,517.94 | Pos Y: -52,658,583,827.34 | Speed: 46,324.5 m/s
Day 49999725 -> Pos X: 44,943,849,977.73 | Pos Y: -43,758,477,253.14 | Speed: 43,757.7 m/s
Day 49999730 -> Pos X: 57,588,158,734.38 | Pos Y: -30,381,549,029.00 | Speed: 41,920.7 m/s
Day 49999735 -> Pos X: 65,009,519,203.36 | Pos Y: -14,222,652,907.17 | Speed: 40,841.0 m/s
Day 49999740 -> Pos X: 66,899,190,274.57 | Pos Y: 3,156,942,048.27 | Speed: 40,523.9 m/s
Day 49999745 -> Pos X: 63,202,389,199.35 | Pos Y: 20,271,990,453.45 | Speed: 40,970.2 m/s
Day 49999750 -> Pos X: 54,087,015,748.09 | Pos Y: 35,636,158,758.68 | Speed: 42,178.9 m/s
Day 49999755 -> Pos X: 39,992,614,118.45 | Pos Y: 47,688,583,329.59 | Speed: 44,142.3 m/s
Day 49999760 -> Pos X: 21,771,971,579.77 | Pos Y: 54,750,524,821.70 | Speed: 46,826.4 m/s
Day 49999765 -> Pos X: 951,330,644.96 | Pos Y: 55,077,549,950.21 | Speed: 50,117.7 m/s
Day 49999770 -> Pos X: -19,901,664,813.84 | Pos Y: 47,163,958,283.00 | Speed: 53,712.4 m/s
Day 49999775 -> Pos X: -36,928,527,069.97 | Pos Y: 30,559,306,611.11 | Speed: 56,955.7 m/s
Day 49999780 -> Pos X: -45,582,816,923.26 | Pos Y: 7,210,482,040.51 | Speed: 58,838.3 m/s
Day 49999785 -> Pos X: -42,841,587,338.33 | Pos Y: -17,889,417,447.32 | Speed: 58,540.5 m/s
Day 49999790 -> Pos X: -29,545,550,712.84 | Pos Y: -38,671,470,108.20 | Speed: 56,204.7 m/s
Day 49999795 -> Pos X: -9,786,250,362.07 | Pos Y: -51,241,467,817.13 | Speed: 52,784.3 m/s
Day 49999800 -> Pos X: 11,826,957,302.65 | Pos Y: -54,843,140,098.34 | Speed: 49,221.9 m/s
Day 49999805 -> Pos X: 31,861,652,468.38 | Pos Y: -50,652,471,996.87 | Speed: 46,068.6 m/s
Day 49999810 -> Pos X: 48,223,321,856.34 | Pos Y: -40,501,286,669.19 | Speed: 43,564.3 m/s
Day 49999815 -> Pos X: 59,766,864,551.81 | Pos Y: -26,251,899,752.84 | Speed: 41,794.0 m/s
Day 49999820 -> Pos X: 65,929,045,009.68 | Pos Y: -9,614,636,943.42 | Speed: 40,781.9 m/s
Day 49999825 -> Pos X: 66,500,432,094.91 | Pos Y: 7,847,250,256.52 | Speed: 40,532.7 m/s
Day 49999830 -> Pos X: 61,515,883,008.62 | Pos Y: 24,651,615,287.34 | Speed: 41,046.7 m/s
Day 49999835 -> Pos X: 51,234,140,549.74 | Pos Y: 39,318,809,281.11 | Speed: 42,323.0 m/s
Day 49999840 -> Pos X: 36,197,543,621.88 | Pos Y: 50,302,292,641.36 | Speed: 44,352.5 m/s
Day 49999845 -> Pos X: 17,384,726,504.06 | Pos Y: 55,957,352,790.27 | Speed: 47,096.9 m/s
Day 49999850 -> Pos X: -3,521,226,204.52 | Pos Y: 54,620,270,592.81 | Speed: 50,431.5 m/s
Day 49999855 -> Pos X: -23,777,066,104.61 | Pos Y: 44,964,117,324.16 | Speed: 54,027.7 m/s
Day 49999860 -> Pos X: -39,408,504,143.43 | Pos Y: 26,880,346,819.19 | Speed: 57,194.0 m/s

Day 49999865 -> Pos X: -46,001,642,774.95 | Pos Y: 2,798,317,608.72 | Speed: 58,906.2 m/s
Day 49999870 -> Pos X: -41,055,809,971.46 | Pos Y: -21,946,484,259.21 | Speed: 58,404.8 m/s
Day 49999875 -> Pos X: -26,059,757,961.58 | Pos Y: -41,423,724,795.23 | Speed: 55,928.6 m/s
Day 49999880 -> Pos X: -5,429,065,190.83 | Pos Y: -52,229,380,349.79 | Speed: 52,462.6 m/s
Day 49999885 -> Pos X: 16,271,286,725.89 | Pos Y: -54,055,504,614.36 | Speed: 48,920.0 m/s
Day 49999890 -> Pos X: 35,799,804,922.91 | Pos Y: -48,332,942,577.78 | Speed: 45,817.9 m/s
Day 49999895 -> Pos X: 51,253,158,591.07 | Pos Y: -37,004,454,267.00 | Speed: 43,376.7 m/s
Day 49999900 -> Pos X: 61,639,195,996.56 | Pos Y: -21,970,882,164.66 | Speed: 41,673.3 m/s
Day 49999905 -> Pos X: 66,513,702,741.89 | Pos Y: -4,950,628,483.05 | Speed: 40,728.9 m/s
Day 49999910 -> Pos X: 65,765,186,969.23 | Pos Y: 12,496,348,780.35 | Speed: 40,547.4 m/s
Day 49999915 -> Pos X: 59,517,563,999.46 | Pos Y: 28,895,499,602.16 | Speed: 41,129.3 m/s
Day 49999920 -> Pos X: 48,120,604,011.91 | Pos Y: 42,778,581,031.62 | Speed: 42,473.0 m/s
Day 49999925 -> Pos X: 32,220,199,199.51 | Pos Y: 52,619,378,016.55 | Speed: 44,568.4 m/s
Day 49999930 -> Pos X: 12,921,504,315.84 | Pos Y: 56,815,965,290.23 | Speed: 47,372.2 m/s
Day 49999935 -> Pos X: -7,937,113,533.96 | Pos Y: 53,798,972,441.76 | Speed: 50,747.6 m/s
Day 49999940 -> Pos X: -27,446,621,552.50 | Pos Y: 42,440,288,788.44 | Speed: 54,339.9 m/s
Day 49999945 -> Pos X: -41,546,871,242.21 | Pos Y: 22,993,858,756.55 | Speed: 57,420.8 m/s
Day 49999950 -> Pos X: -46,011,742,782.11 | Pos Y: -1,633,584,108.60 | Speed: 58,955.9 m/s
Day 49999955 -> Pos X: -38,905,875,089.52 | Pos Y: -25,823,241,273.30 | Speed: 58,253.0 m/s
Day 49999960 -> Pos X: -22,343,022,137.54 | Pos Y: -43,857,860,357.17 | Speed: 55,644.9 m/s
Day 49999965 -> Pos X: -1,000,831,768.81 | Pos Y: -52,847,252,257.17 | Speed: 52,140.5 m/s
Day 49999970 -> Pos X: 20,642,700,826.37 | Pos Y: -52,912,579,446.41 | Speed: 48,621.8 m/s
Day 49999975 -> Pos X: 39,551,949,172.06 | Pos Y: -45,714,124,561.92 | Speed: 45,572.4 m/s
Day 49999980 -> Pos X: 54,016,484,001.17 | Pos Y: -33,287,661,886.41 | Speed: 43,194.9 m/s
Day 49999985 -> Pos X: 63,195,114,592.39 | Pos Y: -17,561,514,585.38 | Speed: 41,558.5 m/s
Day 49999990 -> Pos X: 66,760,418,881.83 | Pos Y: -255,122,562.00 | Speed: 40,681.8 m/s
Day 49999995 -> Pos X: 64,697,298,734.65 | Pos Y: 17,079,943,111.52 | Speed: 40,568.2 m/s